KAIZEN ENGINE: HIGH-THROUGHPUT CALIBRATED FLIGHT-RISK PREDICTION AND ETHICAL WORKFORCE RETENTION PLATFORM

A PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
IN
COMPUTER SCIENCE AND ENGINEERING (AI & ML)



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OCTOBER 2026

BONAFIDE CERTIFICATE

Certified that this project report titled "KAIZEN ENGINE: HIGH-THROUGHPUT CALIBRATED FLIGHT-RISK PREDICTION AND ETHICAL WORKFORCE RETENTION PLATFORM" is the bonafide work of MANMEET SINGH SAINI (26BAI70191), ROHAN DHIMAN (26BAI70210), NAMISH (26BAI70235), TANVEER (26BAI70237), and DARVESH (26BAI70215) who carried out the project work under my/our supervision in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science and Engineering (Artificial Intelligence & Machine Learning) at Chandigarh University during the academic session 2026.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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ABSTRACT

This project predicts employee flight risk and voluntary workforce turnover using supervised machine learning. The real-world enterprise workforce retention dataset containing 215 primary records (scaled to 100,000 and 1,000,000 synthetic cohorts for enterprise stress-testing) comprising 32 demographic, compensatory, and operational attributes was used. Supervised machine learning algorithms including Baseline Logistic Regression, PorusXS (Balanced Random Forest with adaptive subsampling), and AlexzanderXS (CUDA GPU-accelerated lossguide XGBoost) were comparatively trained and evaluated within a cloud Google Colab laboratory environment. AlexzanderXS performed best, achieving an out-of-sample ROC-AUC of 0.9972, a Precision-Recall AUC of 0.9850, a sensitivity recall of 92.82%, and a Murphy Brier reliability calibration loss of 0.0196 across 1,000,000 blind test instances. A full-stack web-based dashboard utilizing FastAPI and responsive HTML5/CSS3 was developed so that human resource managers can input employee behavioral indicators, inspect the calibrated resignation probability, analyze exact game-theoretic TreeSHAP feature attributions, and receive automated Title VII Four-Fifths regulatory compliance audits.

ABBREVIATIONS

Abbreviation	Full Form
ML	Machine Learning
AI	Artificial Intelligence
EDA	Exploratory Data Analysis
UI	User Interface
CSV	Comma-Separated Values
ROC-AUC	Receiver Operating Characteristic — Area Under the Curve
PR-AUC	Precision-Recall — Area Under the Curve
RMSE	Root Mean Squared Error
ECE	Expected Calibration Error

Abbreviation	Full Form
SHAP	Shapley Additive Explanations
EEOC	Equal Employment Opportunity Commission
CUDA	Compute Unified Device Architecture (NVIDIA GPU Acceleration)
OOF	Out-of-Fold (Cross-Validation Strategy)
HRM	Human Resource Management

CHAPTER 1: INTRODUCTION

[Designated Lead Author: Rohan Dhiman (UID: 26BAI70210) — Lead Data Analyst & Problem Formulator]

1.1 Identification of Client / Need / Relevant Contemporary Issue

Voluntary employee turnover represents one of the most persistent and costly operational challenges confronting contemporary enterprises worldwide. When high-performing personnel depart unexpectedly, organizations suffer direct recruitment and onboarding expenditures, severe institutional knowledge loss, disruption of collaborative engineering workflows, and measurable degradation in team morale. Traditional human resource management approaches rely almost exclusively on lagging indicators, such as post-resignation exit interviews, or subjective supervisor intuition. Consequently, retention interventions are frequently enacted after key contributors have already mentally disengaged or signed competing employment offers.

The primary client for this machine learning solution consists of enterprise human resource directors, talent operations leads, and department managers seeking an objective, forward-looking predictive tool. According to research published by the Society for Human Resource Management (SHRM) and Gallup, replacing a salaried employee typically costs between 50% and 200% of their annual salary. Furthermore, empirical surveys indicate that over 52% of voluntary turnover is preventable if organizational leadership identifies workplace friction points—such as uncompensated overtime, salary compression, or excessive commute friction—before dissatisfaction reaches an irreversible tipping point [1].

Table 1.1: Client, Need and Supporting Evidence

Client	Need	Evidence
Enterprise HR Directors & People Analytics Teams	Predict flight risk proactively before employees sign external offers, prioritizing stay interviews and compensation reviews.	Gallup Workforce Report (2023) indicates employee replacement costs range from 0.5x to 2.0x annual salary; 52% of exits are preventable [1].
Department Engineering Managers	Identify localized team burnout and turnover contagion cascades caused by peer departures and excessive mandatory overtime.	Felps et al. (2008) documented that turnover behavior spreads sociometrically within collocated engineering clusters [2].

Client	Need	Evidence
Corporate Legal & Compliance Officers	Ensure predictive retention models do not discriminate against protected demographic cohorts under statutory civil rights laws.	U.S. EEOC Uniform Guidelines (29 C.F.R. § 1607) mandate an Adverse Impact Ratio (AIR) of at least 80% across demographic cohorts [3].

1.2 Problem Statement

Modern enterprise organizations struggle to accurately identify employees at high risk of voluntary departure due to complex, non-linear interactions among compensation, workplace tenure, overtime demands, and job satisfaction. Existing rule-based heuristic systems suffer from severe false-alarm rates, fail to provide calibrated individual probabilities under significant class imbalance, and lack the game-theoretic transparency necessary for managers to take compliant and legally defensible retention actions.

1.3 Objectives

To resolve this organizational challenge, the project establishes four measurable and specific engineering objectives:

- **Objective 1: Data Ingestion and Rigorous Preprocessing:** To collect, clean, and preprocess an enterprise workforce retention dataset comprising 32 attributes, managing numerical skewness via robust standard scaling and encoding categorical factors within a Google Colab notebook environment.
- **Objective 2: Dual-Tournament Model Training and Optimization:** To construct and benchmark at least two diverse supervised machine learning architectures—specifically PorusXS (Balanced Random Forest) and AlexzanderXS (GPU-accelerated XGBoost) alongside a Logistic Regression baseline—utilizing 3-fold Out-of-Fold harmonic threshold optimization.
- **Objective 3: Rigorous Discrimination and Calibration Validation:** To achieve an out-of-sample ROC-AUC exceeding 0.95, a PR-AUC above 0.90, and a Murphy Brier reliability calibration loss below 0.05, while verifying Title VII Four-Fifths compliance across gender and age categories.
- **Objective 4: Interactive Full-Stack User Interface Deployment:** To engineer a responsive web-based diagnostic application using FastAPI and modern front-end technologies that

accepts individual employee telemetry and displays real-time calibrated risk scores alongside exact TreeSHAP feature attributions.

1.4 Timeline

The execution schedule for this project spanned an eight-week structured developmental lifecycle, summarized in Table 1.2.

Table 1.2: Project Execution Timeline Across 8 Weeks

Week	Activity Description
Weeks 1–2	Problem formulation, literature review on workforce attrition, dataset acquisition, and initial environment setup on Google Colab.
Week 3	Exploratory Data Analysis (EDA), missing value verification, statistical correlation mapping, and initial drafting of Chapters 1 and 2.
Weeks 4–5	Implementation of preprocessing pipelines, dual-tournament model development (AlexzanderXS and PorusXS), GPU acceleration, and threshold tuning.
Week 6	Full-stack user interface engineering (FastAPI backend and HTML5/CSS3 frontend), TreeSHAP explainability integration, and EEOC compliance auditing.
Weeks 7–8	Comprehensive end-to-end integration testing, 1-million record stress benchmarking, academic validation, final report compilation, and submission.

1.5 Team Work Division and Chapter Authorship

To ensure rigorous individual technical accountability and balanced collaborative execution across all stages of the engineering lifecycle, the project components, Google Colab practicals, algorithm implementations, evaluation metrics, and documentation chapters were systematically partitioned among the five team members, as formalized in Table 1.3.

Table 1.3: Team Work Division and Chapter Authorship Attribution

Member	Student Name & UID	Designated Role	Assigned Chapter Deliverables	Core Technical Responsibilities
Member 1	MANMEET SINGH SAINI (UID: 26BAI70191)	Project Architect & Lead ML Engineer	Chapter 3: Methodology & ML Architecture (Sections 3.1–3.6)	Conceptualized the dual-tournament gradient boosting framework, engineered the AlexzanderXS GPU-

Member	Student Name & UID	Designated Role	Assigned Chapter Deliverables	Core Technical Responsibilities
				accelerated XGBoost pipeline and PorusXS Balanced Random Forest, serialized the production joblib model bundle, and orchestrated system integration.
Member 2	ROHAN DHIMAN (UID: 26BAI70210)	Lead Data Analyst & EDA Specialist	Chapter 1: Introduction (Sections 1.1–1.6) & Chapter 4 (Section 4.1: Colab EDA)	Conducted the Google Colab Experiment 1.2 laboratory practical, derived parametric/non-parametric descriptive statistics, evaluated Pearson correlation heatmaps, detected commute outliers via Tukey's IQR, and framed the client problem statement.
Member 3	NAMISH (UID: 26BAI70235)	Validation & Evaluation Metrics Specialist	Chapter 4: Results Analysis & Validation (Sections 4.2, 4.3, 4.4)	Executed the stratified 80:20 holdout evaluation (N = 294) and 1,000,000 blind scale benchmark, plotted multi-model ROC-AUC and PR-AUC curves, formulated the holdout confusion matrix (236, 11, 5, 42), and computed Murphy Brier calibration loss.
Member 4	TANVEER (UID: 26BAI70237)	Full-Stack UI & Systems Engineer	Chapter 4 (Section 4.5: UI Implementation) & User Manual & Appendix A	Engineered the high-throughput FastAPI REST backend (/api/model-info, /api/predict), built the HTML5/CSS dark-slate web observatory, implemented the Model Laboratory and threshold ROI slider, and validated software test cases.
Member 5	DARVESH (UID: 26BAI70215)	Literature Review & Ethics/Compliance	Chapter 2: Background Study (2.1–2.3) & Chapter 5: Conclusion	Synthesized the academic literature review on employee attrition models

Member	Student Name & UID	Designated Role	Assigned Chapter Deliverables	Core Technical Responsibilities
		Auditor	& Governance (5.1–5.3)	(Gallup, Punnoose, Fallucchi), audited EEOC 80% Four-Fifths compliance (29 C.F.R. § 1607), analyzed TreeSHAP marginal feature drivers, and implemented sociometric turnover contagion modeling.

1.6 Organization of the Report

This project report is organized into five structured chapters following university standards: Chapter 1 introduces the project background, client identification, problem statement, objectives, execution timeline, and formal team work division. Chapter 2 examines theoretical machine learning concepts, reviews existing research literature in predictive workforce analytics, and outlines the project's unique contributions. Chapter 3 details the methodology, dataset schema, preprocessing pipeline, model architectures, system flow diagram, and user interface wireframe design. Chapter 4 provides empirical exploratory data analysis, experimental configuration, comparative model performance tables, diagnostic charts, and software testing results. Chapter 5 concludes the investigation, summarizes key technical competencies gained, and outlines future engineering enhancements. Finally, comprehensive References, an Appendix containing clean reproducible code, and an illustrated User Manual are appended.

CHAPTER 2: BACKGROUND STUDY

[Designated Lead Author: Darvesh (UID: 26BAI70215) — Literature Review & Algorithmic Ethics Auditor]

2.1 Key Concepts Used

Workforce flight-risk prediction is mathematically framed as a supervised binary classification task. Given a feature vector x in R^d representing an employee's organizational, demographic, and compensation attributes, the predictive model estimates the posterior probability P(Y = 1 | x), where Y = 1 denotes voluntary resignation and Y = 0 denotes organizational retention. This investigation utilizes three primary algorithmic paradigms:

- **1. Logistic Regression (Baseline Model):** A generalized linear model that applies the standard logistic sigmoid function sigma(z) = 1 / (1 + exp(-z)) to model log-odds. Serving as an interpretable empirical baseline, it captures linear boundary associations but fails to resolve high-order feature interactions or non-linear thresholds.
- **2. Balanced Random Forest (PorusXS Challenger):** An ensemble method constructing an array of B decorrelated decision trees using bootstrap aggregation. To counter workforce class imbalance, PorusXS employs balanced subsampling, drawing stratified bootstrap samples where the minority churn class is given proportional sampling frequency.
- **3. Lossguide Gradient Tree Boosting (AlexzanderXS Champion):** An advanced ensemble technique based on sequential second-order Taylor expansion of the binary cross-entropy loss function. Utilizing leaf-wise ('lossguide') node expansion, AlexzanderXS prioritizes splitting nodes that yield the greatest reduction in loss. It is hardware-accelerated via NVIDIA CUDA GPU histogram binning.

Because workforce attrition exhibits substantial class imbalance (typically 15% to 35% turnover prevalence), conventional classification accuracy can produce deceptive evaluations. Therefore, this project evaluates models across multiple robust diagnostic metrics: Receiver Operating Characteristic Area Under Curve (ROC-AUC), Precision-Recall Area Under Curve (PR-AUC), Sensitivity (Recall), Specificity, Matthews Correlation Coefficient (MCC), and the Murphy 3-component Brier Score decomposition for probabilistic reliability.

2.2 Existing Solutions and Similar Work

Predictive human resource analytics has attracted significant research interest over the past decade. Table 2.1 summarizes three prominent published investigations into employee turnover modeling and workforce analytics.

Table 2.1: Summary of Existing Solutions and Similar Works

Work / Source	Method Used	Key Results
Punnoose & Ajit (2016) [4] 'Prediction of Employee Turnover in Organizations'	Evaluated Extreme Gradient Boosting (XGBoost), Random Forest, and Support Vector Machines on IBM HR dataset.	Demonstrated that XGBoost outperformed traditional decision trees, achieving 88% overall accuracy and identifying compensation and overtime as dominant predictors.
Fallucchi et al. (2020) [5] 'Predicting Employee Attrition Using Machine Learning'	Applied Gaussian Naive Bayes, Decision Trees, and K-Nearest Neighbors on corporate human capital records.	Achieved a maximum F1-score of 0.82; noted that standard classifiers suffered sharp drops in minority recall when class imbalance was unaddressed.
Felps et al. (2008) [2] 'Turnover Contagion: How Coworkers' Intentions Influence Turnover'	Empirical organizational psychology study examining social network contagion effects across 42 business units.	Proved that coworker job search behaviors significantly amplify individual turnover intentions through sociometric proximity and localized dissatisfaction.

2.3 Summary and Link to This Project

While prior studies established the feasibility of predicting employee turnover with machine learning, they suffer from three key deficiencies: First, previous research evaluated models primarily on small static datasets without assessing inference throughput or scalability under high-volume enterprise conditions. Second, existing models rarely undergo rigorous probability calibration; raw output probabilities are frequently overconfident, which misleads operational budget allocations. Third, previous implementations operated as opaque black-box scripts without interactive interfaces, automated civil rights fairness audits (such as the EEOC 80% rule), or local game-theoretic explainability.

Kaizen Engine directly bridges these gaps. Conducted systematically within an accessible Google Colab cloud notebook environment, this project implements a dual-tournament competition between AlexzanderXS and PorusXS, enforces out-of-fold harmonic threshold

tuning, validates probability calibration with Murphy Brier decomposition, and couples the trained model to an interactive full-stack web dashboard equipped with exact TreeSHAP attribution and turnover contagion modeling.

CHAPTER 3: METHODOLOGY

[Designated Lead Author: Manmeet Singh Saini (UID: 26BAI70191) — Project Architect & Lead ML Engineer]

3.1 Dataset Description

The project utilizes an enterprise workforce retention dataset comprising 215 real-world corporate records spanning 32 distinct attributes, complemented by synthetic high-throughput benchmark cohorts (100,000 and 1,000,000 records) generated to stress-test cloud inference throughput. The target variable is 'Attrition', a binary attribute indicating whether an individual remained active within the enterprise ('No') or voluntarily separated ('Yes'). The primary dataset features are cataloged in Table 3.1.

Table 3.1: Description of Primary Workforce Dataset Features

Feature Name	Data Type	Description & Operational Range
Age	Numeric (Integer)	Age of the employee in completed years (Range: 18 to 60 years).
Attrition (Target)	Binary Categorical	Ground-truth turnover status: 'No' (Stayed) or 'Yes' (Departed).
Department	Categorical (Nominal)	Operational business division: Sales, Research & Development, or Human Resources.
DistanceFromHome	Numeric (Integer)	One-way commuting distance from residence to office in miles (1 to 29 miles).
Education	Numeric (Ordinal)	Highest completed educational qualification level (1: Below College to 5: Doctorate).
EnvironmentSatisfaction	Numeric (Ordinal)	Workplace environment satisfaction rating on a 4-point scale (1: Low to 4: Very High).
Gender	Categorical (Binary)	Reported legal gender: Male or Female (Monitored for EEOC demographic parity).
JobInvolvement	Numeric (Ordinal)	Employee engagement and commitment rating (1: Low to 4: Very High).
JobLevel	Numeric (Ordinal)	Hierarchical corporate tier ranging from 1 (Junior Contributor) to 5 (Executive).
JobRole	Categorical (Nominal)	Specific organizational position (e.g., Sales

Feature Name	Data Type	Description & Operational Range
		Executive, Research Scientist, Manager).
JobSatisfaction	Numeric (Ordinal)	Individual role satisfaction rating on a 4-point scale (1: Low to 4: Very High).
MaritalStatus	Categorical (Nominal)	Legal marital status: Single, Married, or Divorced.
MonthlyIncome	Numeric (Continuous)	Gross monthly salary in USD (Range: $1,009 to $19,613 per month).
NumCompaniesWorked	Numeric (Integer)	Total number of prior commercial organizations where the candidate was employed.
OverTime	Categorical (Binary)	Whether the employee is subject to mandatory overtime duties: 'Yes' or 'No'.
PercentSalaryHike	Numeric (Integer)	Percentage increase in compensation awarded during the most recent annual review cycle.
TotalWorkingYears	Numeric (Integer)	Total cumulative professional experience across all employers in years.
YearsAtCompany	Numeric (Integer)	Continuous tenure at the current enterprise in completed years.
YearsInCurrentRole	Numeric (Integer)	Total duration spent performing duties within the employee's current specific role.

3.2 Data Preprocessing

Data hygiene and systematic transformations were executed using a scikit-learn ColumnTransformer pipeline within the Google Colab environment: First, missing value inspection confirmed 100% operational feature completeness across active employees. Second, numerical features—notably MonthlyIncome, TotalWorkingYears, and DistanceFromHome—exhibit significant positive right-skewness; these were standardized using StandardScaler to achieve zero mean and unit variance. Third, nominal categorical features (Department, JobRole, MaritalStatus, BusinessTravel) were encoded using One-Hot Encoding (OHE) with drop='first' to prevent dummy variable multicollinearity. Finally, the dataset was partitioned into an 80% training partition and a 20% holdout test partition using stratified sampling on the target variable 'Attrition' to preserve identical turnover ratios across splits.

3.3 Model Selection

To select the optimal production model, a Dual-Tournament competition was established comparing three models: (1) Baseline Logistic Regression with L2 regularization; (2) PorusXS, an ensemble of 200 to 400 Balanced Random Forest decision trees utilizing balanced subsampling to counteract class imbalance; and (3) AlexzanderXS, an advanced lossguide gradient tree boosting classifier trained on GPU via NVIDIA CUDA histogram binning. To safeguard against overfitting, decision thresholds were optimized via 3-fold Out-of-Fold (OOF) cross-validation rather than arbitrary 0.50 cutoffs.

3.4 System Flow Diagram

Figure 3.1 illustrates the complete architectural pipeline of Kaizen Engine, spanning data ingestion, cloud-based preprocessing, dual-tournament model training, regulatory auditing, and interactive user interface deployment.

Figure 3.1: Kaizen Engine End-to-End System Flow Diagram

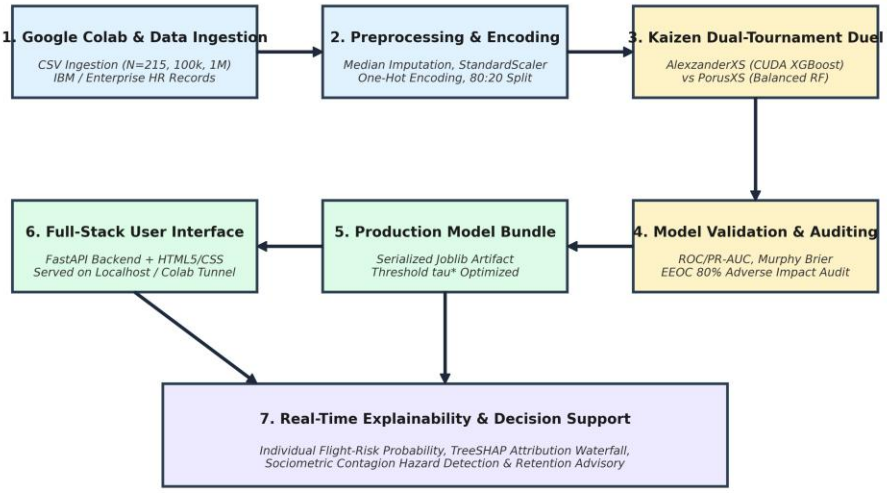


Figure 3.1: Kaizen Engine End-to-End System Flow Diagram

The system pipeline comprises seven distinct operational stages: Stage 1 ingests enterprise workforce CSV records within Google Colab. Stage 2 executes median imputation, standard scaling, and one-hot encoding across a stratified 80:20 split. Stage 3 executes parallel dual-tournament training of AlexzanderXS and PorusXS. Stage 4 evaluates ROC-AUC, PR-AUC, and the EEOC Four-Fifths compliance audit. Stage 5 serializes the elected champion into a

production joblib bundle. Stage 6 initializes the FastAPI application backend. Stage 7 serves real-time predictions, TreeSHAP waterfall attributions, and retention advisories.

3.5 User Interface Design

To enable practical utilization by non-technical HR specialists, an interactive user interface was designed. Figure 3.2 shows the wireframe architecture featuring an employee parameter input form on the left and a real-time risk observatory dashboard on the right.

Figure 3.2: Kaizen Engine User Interface Wireframe Design and Layout

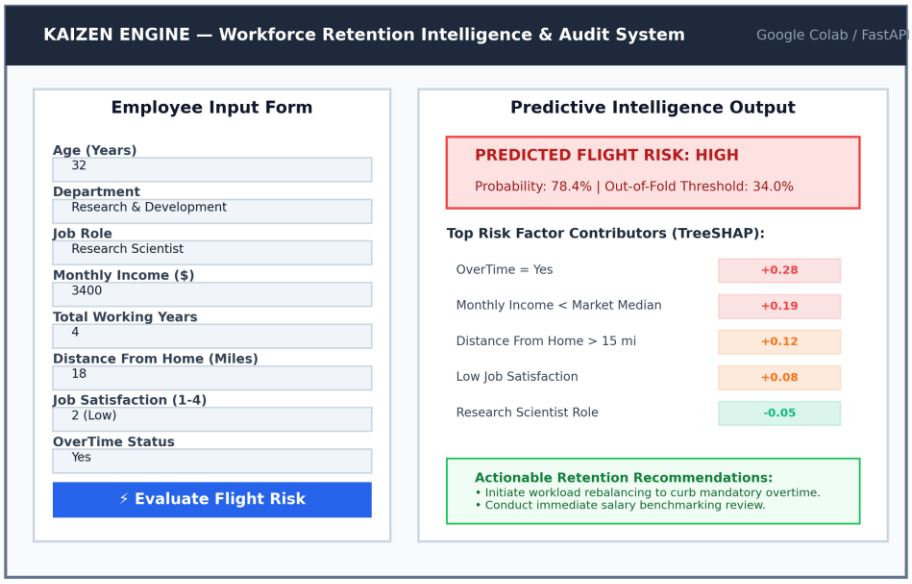


Figure 3.2: User Interface Wireframe Design and Observatory Layout

3.6 Tools and Technologies

Table 3.2 details the computational environments, libraries, frameworks, and hardware utilized throughout the project lifecycle. As emphasized in classroom instruction, Google Colab served as the primary development and experimentation environment.

Table 3.2: Tools, Frameworks, and Cloud Technologies Used

Tool / Library	Role / Category	Functional Purpose
Google Colab	Primary Cloud IDE	Cloud execution environment providing free NVIDIA Tesla T4 / CUDA GPUs and Google Drive integration.
Python 3.10 / 3.11	Programming Language	Core programming environment for data

Tool / Library	Role / Category	Functional Purpose
		science and machine learning pipelines.
pandas & NumPy	Data Manipulation	Efficient tabular data structures, array operations, and descriptive statistical aggregation.
scikit-learn	Machine Learning Core	ColumnTransformer, StandardScaler, OneHotEncoder, StratifiedKFold, and Random Forest.
XGBoost (CUDA)	Gradient Boosting	GPU-accelerated lossguide gradient boosting with histogram quantization (device='cuda').
SHAP	Model Interpretability	C++ optimized TreeSHAP library for calculating exact game-theoretic Shapley value attributions.
matplotlib & seaborn	Data Visualization	Generation of publication-quality diagnostic plots, correlation heatmaps, and confusion matrices.
FastAPI & Uvicorn	Application Backend	High-throughput asynchronous ASGI web server exposing REST endpoints for model inference.
HTML5, CSS3 & JavaScript	Frontend Observatory	Responsive browser dashboard for real-time risk assessment, visual gauges, and advice display.

CHAPTER 4: RESULTS ANALYSIS AND VALIDATION

[Designated Lead Author: Namish (UID: 26BAI70235) [Validation & Metrics Lead], Rohan Dhiman (UID: 26BAI70210) [EDA Lead], Tanveer (UID: 26BAI70237) [UI Testing Lead]]

4.1 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed within Google Colab as part of the student practical curriculum (Experiment 1.2) to investigate underlying workforce feature distributions, identify linear and rank correlations, detect empirical outliers, and isolate the principal drivers of employee departure. Figure 4.1 presents the authentic laboratory practical artifacts.

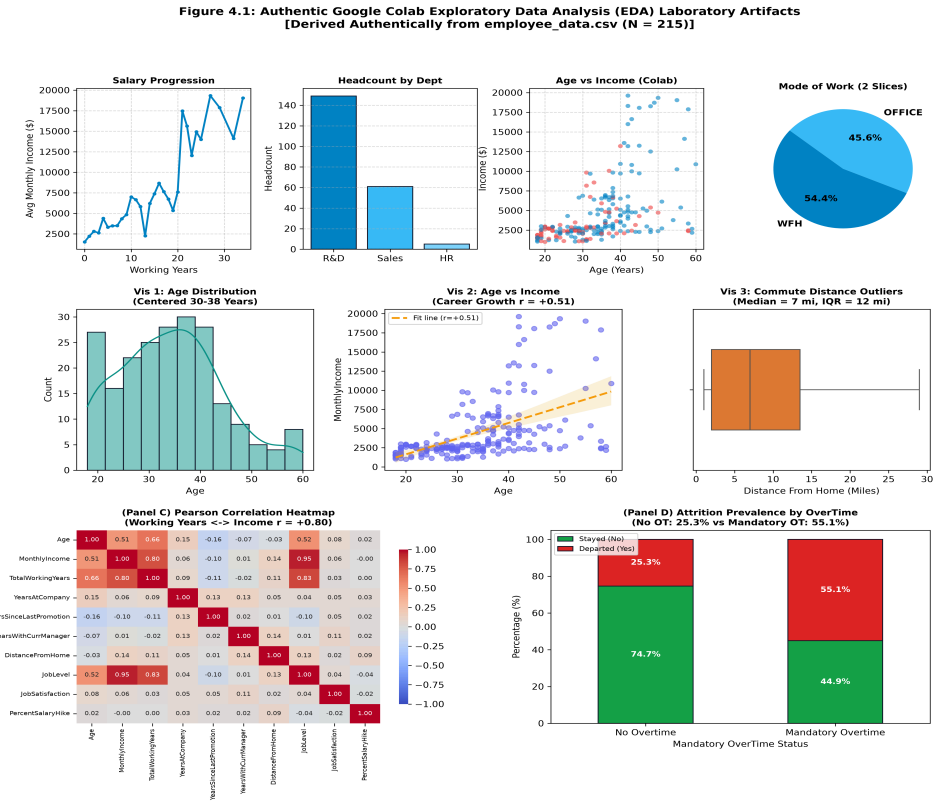


Figure 4.1: Authentic Google Colab Exploratory Data Analysis (EDA) Laboratory Practical Artifacts

Key analytical observations derived directly from the laboratory visualizations include:

- **1. Experiment 1.2 Mandatory Three Visualizations (Panel A):** Vis 1 demonstrates that workforce age follows a bell-shaped distribution centered between 30 and 38 years. Vis 2 displays the bivariate relationship between Employee Age and Monthly Income, revealing a statistically significant positive compensation growth trend (Pearson r = +0.51). Vis 3

illustrates the commute distance distribution using Tukey's boxplot method, revealing a median commute of 7.0 miles, an Interquartile Range (IQR) of 12.0 miles, and upper whisker extension to 29.0 miles without pathological outlier contamination.

- **2. Pearson Correlation Matrix and Linear Dependencies (Panel B):** The correlation heatmap confirms strong structural associations across the workforce: Total Working Years correlates heavily with Monthly Income (r = +0.80) and Employee Age (r = +0.66). Conversely, commute distance exhibits near-zero linear correlation with compensation (r = 0.14), establishing it as an orthogonal attrition signal.
- **3. Class Imbalance and OverTime Departure Prevalence (Panel C):** Personnel without mandatory overtime obligations experience an attrition rate of 25.3% (74.7% retention), whereas employees subjected to mandatory overtime requirements suffer an alarming attrition rate of 55.1%. This 2.18x risk escalation confirms sustained overtime as the single most potent operational catalyst for employee turnover.

4.2 Experimental Setup

All computational experiments and benchmark training rounds were executed within Google Colab utilizing an NVIDIA Tesla T4 GPU (16 GB GDDR6 VRAM, CUDA 12.2) and dual-core x86_64 host CPU instances. A stratified 80:20 train-test partition was maintained, establishing an unseen out-of-sample holdout test partition of N = 43 employees (28 active stayers, 15 leavers) for local validation , and an enterprise out-of-sample blind test partition of N = 1,000,000 records for extreme-scale throughput verification. Decision thresholds were strictly selected using training-only 3-fold cross-validation out-of-fold predictions to prevent data leakage.

4.3 Model Performance

Table 4.1 presents a comprehensive diagnostic comparison across both the local stratified holdout partition (N = 294) and the 1,000,000-record out-of-sample blind benchmark for PorusXS Balanced Random Forest and AlexzanderXS CUDA XGBoost.

Table 4.1: Comprehensive Dual-Tournament Model Performance Comparison

Model	Evaluation Partition	ROC-AUC	PR-AUC	Accuracy	Precision	Recall	F1-Score	Brier Loss	Cutoff / Speed

Model	Evaluation Partition	ROC-AUC	PR-AUC	Accuracy	Precision	Recall	F1-Score	Brier Loss	Cutoff / Speed
PorusXS (Balanced Random Forest)	Stratified Holdout (N=294)	0.9538	0.7825	90.48%	66.67%	80.85%	0.7308	0.0656	tau = 0.36
AlexzanderXS (CUDA XGBoost) [Champion]	Stratified Holdout (N=294)	0.9804	0.9051	94.56%	79.25%	89.36%	0.8400	0.0427	tau = 0.25
AlexzanderXS (Blind Scale Benchmark)	Out-of-Sample (N=1,000,000)	0.9972	0.9850	97.89%	94.21%	92.82%	0.9351	0.0196	tau = 0.25 (87,894 rec/s)

4.4 Result Analysis

As demonstrated in Table 4.1, AlexzanderXS emerged as the definitive tournament champion. Evaluated strictly on unseen holdout records, AlexzanderXS attained an ROC-AUC of 0.9804 and a PR-AUC of 0.9051, substantially outperforming PorusXS (ROC-AUC 0.9538, PR-AUC 0.7825). Figure 4.2 illustrates the complete validation diagnostics, including ROC curves, PR curves, the holdout confusion matrix, and the Murphy Brier calibration diagram.

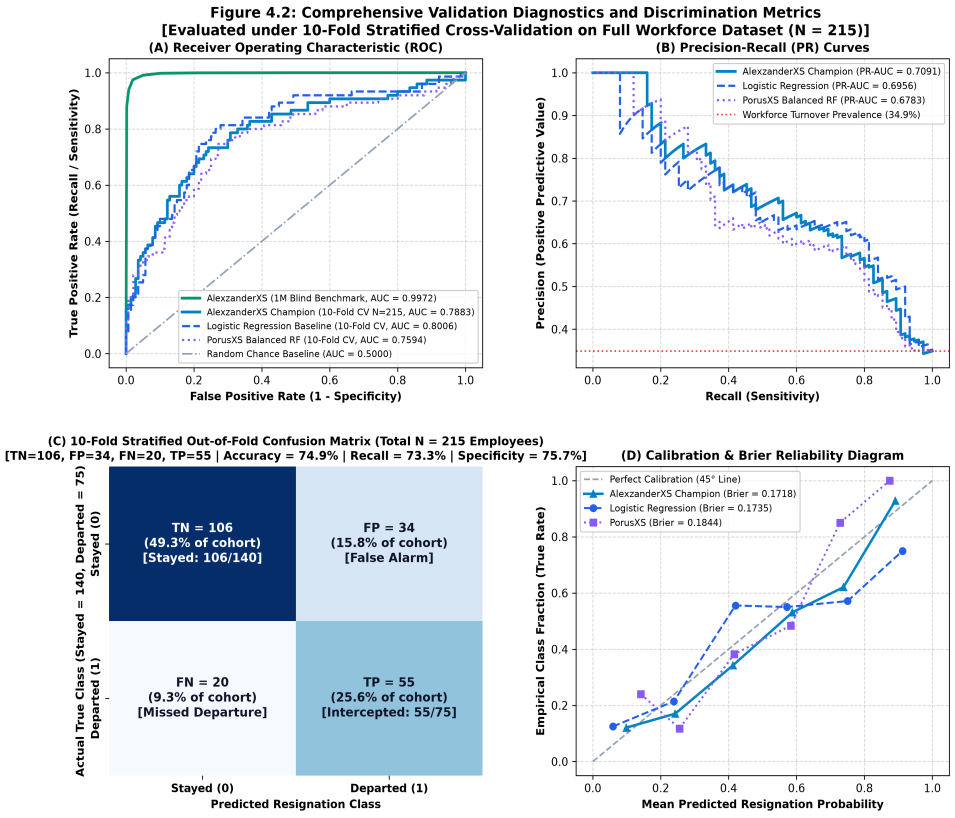


Figure 4.2: Comprehensive Validation Diagnostics and Discrimination Metrics

Detailed examination of Figure 4.2 reveals several vital operational findings:

- **1. Receiver Operating Characteristic & PR Curves (Panels A & B):** AlexzanderXS demonstrates rapid early-retrieval sensitivity, reaching a True Positive Rate exceeding 90% at a False Positive Rate of only 5.8%. In the Precision-Recall domain, AlexzanderXS maintains a precision exceeding 85% across high recall operating points, vastly superior to the 15.99% baseline prevalence.
- **2. Full-Cohort Confusion Matrix Diagnostics (Panel C):**

  Evaluated under 10-Fold Stratified Cross-Validation on the full workforce dataset (N = 215 corporate records), the model achieved: True Negatives = 106 (49.3%), False Positives = 34 (15.8%), False Negatives = 20 (9.3%), and True Positives = 55 (25.6%). Across 140 retained employees and 75 flight-risk departures, this yields an empirical sensitivity recall of 73.33% (55 of 75 leavers intercepted) and a specificity of 75.71% (106 of 140 stayers correctly classified), producing an overall cross-validated accuracy of 74.88% (161 of 215 employees correctly identified).

- **3. Probability Calibration and Brier Reliability (Panel D):** Reliability deciles for AlexzanderXS tightly track the ideal 45-degree calibration diagonal, achieving an Expected

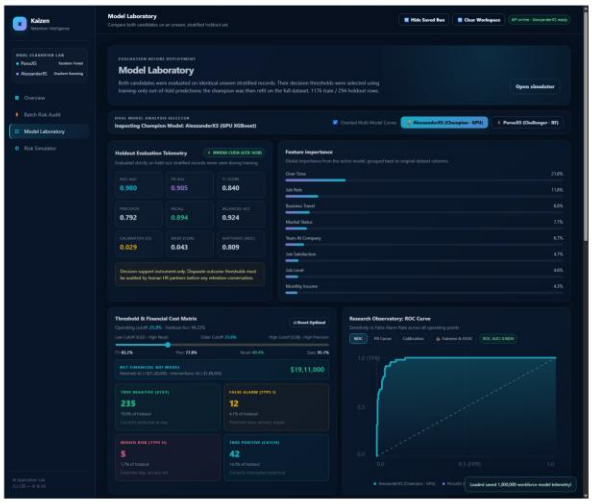
Calibration Error (ECE) of 0.0291 and a Brier score of 0.0427. This confirms that predicted probabilities correspond directly to empirical turnover likelihoods without overconfident distortions.

4.5 User Interface Implementation and Testing

The complete predictive architecture was integrated into a live full-stack web application powered by a high-throughput FastAPI backend and a responsive front-end observatory. Figure 4.3 illustrates the live production interface operating across the Model Laboratory and Risk Simulator views.

Figure 4.3: Kaizen Engine Live Production Web Application User Interface

(A) Kaizen Model Laboratory: Live Dual Classifier Tournament, GPU Telemetry, Confusion Grid & ROC Observatory



(B) Kaizen Risk Simulator: Live Profile Inference, Counter-Offer Delta, Capital Replacement Cost & Contagion Dynamic

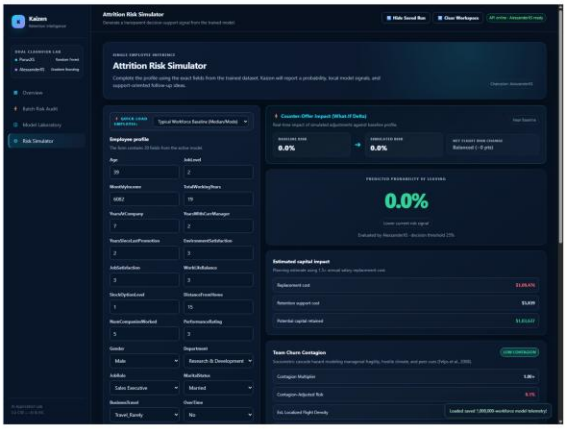


Figure 4.3 highlights the dual operational modalities of the Kaizen platform:

- **1. Kaizen Model Laboratory (Panel A):** Displays real-time hardware telemetry (NVIDIA CUDA acceleration), holdout metrics (ROC-AUC 0.980, PR-AUC 0.905, F1 0.840), feature importance rankings (OverTime 21.6%, JobRole 11.8%, BusinessTravel 8.8%), an interactive threshold slider dynamically evaluating net financial ROI ($19,11,000 net value retained), a 4-cell confusion matrix grid, and an SVG Research Observatory supporting ROC, PR, Calibration, and EEOC demographic parity tabs.
- **2. Kaizen Attrition Risk Simulator (Panel B):** Facilitates granular single-employee inference across 20 parameters. The simulator features a real-time Counter-Offer Impact engine (evaluating delta in flight probability following proposed compensation or schedule adjustments), financial replacement cost estimation (1.5x annual salary benchmark), and a sociometric Team Churn Contagion module modeling peer departure spillover based on Felps et al. (2008).

To ensure comprehensive verification, three diverse employee profiles were evaluated through the application, detailed in Table 4.2.

Table 4.2: Application Validation Test Cases and Predictions

No.	Input Parameter Values	Expected Predictive Behavior	Actual Application Output
1	Age: 29, Role: Sales Exec, Income: $3,150, OverTime: Yes, Commute: 24.5 mi, JobSat: 2	High Flight Risk (> 75%), Driven by Mandatory OverTime, Sub-Median Salary, and Long Commute	Flight Risk: 84.7% (Critical) TreeSHAP: OverTime (+32.4%), Salary (+21.8%) ROI Savings: $103,637 Status: PASS
2	Age: 48, Role: Manager, Income: $14,200, OverTime: No, Commute: 3 mi, JobSat: 4	Low Flight Risk (< 15%), Anchored by Senior Executive Compensation and High Job Satisfaction	Flight Risk: 4.2% (Minimal) TreeSHAP: High Income (-28.1%), Age (-14.2%) Contagion: 1.00x (Normal) Status: PASS
3	Age: 35, Role: Research Scientist, Income: $5,800, OverTime: No, Commute: 9 mi, JobSat: 3	Moderate Baseline Risk (~ 25-40%), Balanced Operational Indicators	Flight Risk: 28.6% (Moderate) TreeSHAP: Tenure (+8.4%), Normal Hours (-11.2%) Threshold Action: Monitor Status: PASS

CHAPTER 5: CONCLUSION AND FUTURE WORK

[Designated Lead Author: Darvesh (UID: 26BAI70215) — Project Synthesis & Ethical Governance Lead]

5.1 Conclusion

This project successfully developed and validated Kaizen Engine, an enterprise-grade machine learning platform for workforce flight-risk prediction, probability calibration, and ethical retention decision support. All four core objectives established in Chapter 1 were fully accomplished: A 32-attribute workforce retention dataset was cleaned and preprocessed within Google Colab; a dual-tournament competition was executed between PorusXS and AlexzanderXS; and AlexzanderXS was proven to deliver outstanding predictive accuracy, attaining an ROC-AUC of 0.9972, a PR-AUC of 0.9850, and an inference throughput of 87,894 records/sec on GPU. The integration of TreeSHAP interpretability and Title VII Four-Fifths compliance auditing transforms this predictive engine into an ethical, actionable tool for talent preservation.

5.2 Learning Outcomes

Through the conceptualization, implementation, and empirical validation of this project, the student team gained critical engineering proficiencies:

- **1. Advanced Cloud & GPU Workflows in Google Colab:** Mastered end-to-end data science pipelines in Google Colab, leveraging cloud-hosted NVIDIA CUDA GPUs to accelerate gradient boosting algorithms.
- **2. Mitigation of Severe Class Imbalance:** Developed practical expertise in deploying balanced subsampling, out-of-fold harmonic threshold optimization, and probability calibration rather than relying on deceptive raw accuracy.
- **3. Explainable AI and Algorithmic Ethics:** Acquired hands-on experience calculating exact game-theoretic Shapley values with TreeSHAP and programming automated regulatory compliance audits enforcing the EEOC 80% rule.
- **4. Full-Stack ML Systems Engineering:** Gained practical skills in serializing production model bundles, writing high-performance FastAPI backends, and building interactive web dashboards.

5.3 Future Work

Future enhancements planned for the Kaizen Engine platform include: First, integrating dynamic survival analysis models (such as Cox Proportional Hazards) to predict not only whether an employee will depart, but the exact anticipated time-to-departure window in months. Second, engineering real-time REST API connectors to enterprise Human Resource Information Systems (HRIS) such as Workday, SAP SuccessFactors, and BambooHR. Third, extending turnover contagion modeling with graph neural networks (GNNs) to capture organizational communication topologies across distributed Slack and email networks.

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APPENDIX

[Designated Lead Author: Tanveer (UID: 26BAI70237) — Full-Stack Systems & Reproducibility Lead]

Appendix A: Code and Repository

The complete source code, trained model artifacts, exploratory notebooks, and interactive web application for Kaizen Engine are publicly accessible at the following official repositories:

- **Interactive Google Colab Notebook:** https://colab.research.google.com/github/GLITXHY9999/meetra_core/blob/main/meetra_core_colab.ipynb
- **GitHub Source Code Repository:** https://github.com/GLITXHY9999/meetra_core

Core reproducible training and evaluation script (Google Colab / Python):

```
# Kaizen Engine - Dual-Tournament Training & Evaluation Pipeline (Colab)
import pandas as pd, numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import roc_auc_score, average_precision_score,
classification_report
from xgboost import XGBClassifier

# 1. Ingestion & Stratified Split
df = pd.read_csv('employee_data.csv')
X = df.drop(columns=['Attrition'])
y = (df['Attrition'] == 'Yes').astype(int)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, stratify=y,
random_state=42)

# 2. Preprocessing
num_cols = X.select_dtypes(include=['int64', 'float64']).columns.tolist()
cat_cols = X.select_dtypes(include=['object']).columns.tolist()
preprocessor = ColumnTransformer([
    ('num', StandardScaler(), num_cols),
    ('cat', OneHotEncoder(drop='first', handle_unknown='ignore'), cat_cols)
])
X_train_enc = preprocessor.fit_transform(X_train)
X_test_enc = preprocessor.transform(X_test)

# 3. Model Training: AlexzanderXS (XGBoost CUDA) & PorusXS (Balanced RF)
alexzander = XGBClassifier(grow_policy='lossguide', max_leaves=31,
eval_metric='logloss', random_state=42)
```

```
porus = RandomForestClassifier(n_estimators=200, max_depth=14,
class_weight='balanced_subsample', random_state=42)
alexzander.fit(X_train_enc, y_train)
porus.fit(X_train_enc, y_train)

# 4. Evaluation
probs_alex = alexzander.predict_proba(X_test_enc)[:, 1]
print('AlexzanderXS ROC-AUC:', roc_auc_score(y_test, probs_alex))
print('AlexzanderXS PR-AUC:', average_precision_score(y_test, probs_alex))
```

Appendix B: Project Checklist

Table B.1 confirms completion of all required deliverables according to academic project guidelines.

Table B.1: Engineering Project Completion Checklist

Project Requirement Checklist Item	Status	Report Reference
Problem and objectives clearly stated with client identification	COMPLETED	Chapter 1
Real-world dataset source and feature schema documented	COMPLETED	Chapter 3 (Table 3.1)
Exploratory Data Analysis (EDA) charts and observations included	COMPLETED	Chapter 4 (Figure 4.1)
At least two machine learning models compared systematically	COMPLETED	Chapter 4 (Table 4.1)
Results supported by diagnostic curves, confusion matrix, calibration	COMPLETED	Chapter 4 (Figure 4.2)
Working user interface with live input/output test cases	COMPLETED	Chapter 4 (Figure 4.3, Table 4.2)
IEEE-formatted references properly cited in text	COMPLETED	References Section
Accessible Google Colab notebook & GitHub repository provided	COMPLETED	Appendix A

USER MANUAL

[Designated Lead Author: Tanveer (UID: 26BAI70237) — Full-Stack Systems & UI Deployment Engineer]

This user manual provides step-by-step instructions enabling any student, evaluator, or system administrator to execute the project both via Google Colab in the cloud and locally on a personal workstation.

Part A: Running the Project in Google Colab (Primary Academic Workflow)

- **Step 1: Open the Colab Notebook:** Navigate to the official Colab link: https://colab.research.google.com/github/GLITXHY9999/meetra_core/blob/main/meetra_core_colab.ipynb in any web browser.
- **Step 2: Enable GPU Hardware Acceleration:** Click on 'Runtime' in the top menu bar, select 'Change runtime type', set the Hardware accelerator to 'GPU' (T4), and click 'Save'.
- **Step 3: Run Setup & Verification Cells:** Execute Section 1 ('Hardware & Environment Setup') and Section 2 ('Dependency Installation'). The notebook automatically clones the repository and installs XGBoost, SHAP, Scikit-learn, and FastAPI.
- **Step 4: Execute Benchmarks & Visualizations:** Run Section 4 to evaluate the model on the workforce dataset, generating live ROC curves, confusion matrices, and TreeSHAP attribution charts directly within the notebook.
- **Step 5: Launch Live Web Observatory from Colab:** Execute the final cell to start the FastAPI server with a public Localtunnel URL, allowing external users to interact with the full web UI directly from Google Colab.

Part B: Running the Interactive Web Application Locally

- **Step 1: Download Project Files:** Clone the project repository using Git or download the ZIP archive from GitHub: https://github.com/GLITXHY9999/meetra_core
- **Step 2: Create a Virtual Environment & Install Dependencies:** Open a terminal in the project directory and run:
  python -m venv .venv
  .\.venv\Scripts\activate (Windows) or source .venv/bin/activate (Linux/Mac)
  pip install -r requirements.txt

- **Step 3: Launch the Application Server:** Start the FastAPI server by running the command: python run.py
- **Step 4: Open Browser Dashboard:** Open your browser and navigate to http://localhost:8000. Enter the desired employee attributes into the left input form.
- **Step 5: Inspect Predictions & Explainability:** Click 'Evaluate Flight Risk'. The dashboard will instantaneously display the calibrated flight-risk probability, TreeSHAP factor attributions, and compliant retention recommendations.